

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/254322444>

Student-Made Games to Learn the History of Mathematics

Article in PRIMUS · August 2011

DOI: 10.1080/10511970903521651

CITATIONS

6

READS

519

2 authors:



[Mary Ann Huntley](#)

Cornell University

25 PUBLICATIONS 578 CITATIONS

[SEE PROFILE](#)

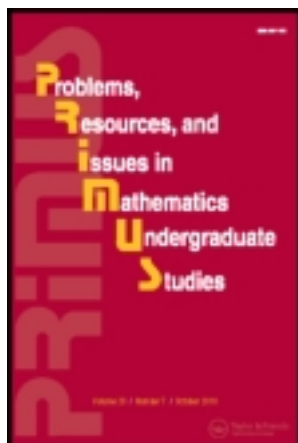


[Alfinio Flores](#)

University of Delaware

108 PUBLICATIONS 772 CITATIONS

[SEE PROFILE](#)



PRIMUS

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/upri20>

Student-Made Games to Learn the History of Mathematics

Mary Ann Huntley & Alfinio Flores

Available online: 17 Jul 2011

To cite this article: Mary Ann Huntley & Alfinio Flores (2011): Student-Made Games to Learn the History of Mathematics, PRIMUS, 21:6, 567-576

To link to this article: <http://dx.doi.org/10.1080/10511970903521651>

PLEASE SCROLL DOWN FOR ARTICLE

Full terms and conditions of use: <http://www.tandfonline.com/page/terms-and-conditions>

This article may be used for research, teaching and private study purposes. Any substantial or systematic reproduction, re-distribution, re-selling, loan, sub-licensing, systematic supply or distribution in any form to anyone is expressly forbidden.

The publisher does not give any warranty express or implied or make any representation that the contents will be complete or accurate or up to date. The accuracy of any instructions, formulae and drug doses should be independently verified with primary sources. The publisher shall not be liable for any loss, actions, claims, proceedings, demand or costs or damages whatsoever or howsoever caused arising directly or indirectly in connection with or arising out of the use of this material.

Student-Made Games to Learn the History of Mathematics

Mary Ann Huntley and Alfinio Flores

Abstract: Prospective secondary mathematics teachers design their own adaptations of popular board and computer games to learn the history of mathematics.

Keywords: Mathematics games, history of mathematics, meaningful practice, motivation, alternative assessment.

1. INTRODUCTION

There is a long tradition of using games effectively to teach mathematics at the high school level [7]. Games have also been used to teach mathematics at the college level [3]. Research has documented the effectiveness of games in the learning of mathematics [2, 8]. In most instances, games are designed by the teacher, but students can also use their own creativity to adapt existing games or invent new games to learn mathematics [4]. In the same way, students can adapt or invent games to learn the history of mathematics.

In this article we describe how college students, most of whom are prospective secondary mathematics teachers, designed and used their own games to learn the history of mathematics. We begin the article by describing some of the games students designed and used, and follow this with a discussion of factors for successful use of games in learning mathematics. We conclude with some of our own reflections and a reflection from a student.

2. GAMES DESIGNED BY STUDENTS

The course on the history of mathematics at the University of Delaware is specially designed to prepare undergraduate secondary mathematics education majors to become better mathematics teachers by enhancing their knowledge

Address correspondence to Mary Ann Huntley, Mathematics Department, Cornell University, Ithaca, NY 14853, USA. E-mail: huntley@math.cornell.edu

of mathematical content and ways of thinking about this content vis-à-vis their future teaching [6]. In this course each student takes lead responsibility for one of the mathematical topics explored throughout the semester. Through their assigned readings, writing assignments, and in-class presentations to their peers, students learn about the historical aspects of all the topics, and investigate deeply the historical aspects of their assigned mathematical topic.

One of their assignments involves preparing and delivering a 45-minute mathematics lesson on an aspect of their topic to their classmates. Approximately one-third of the students choose to use games as a mechanism for incorporating the history of mathematics into their lessons. After their presentation, students write a 3- to 4-page paper summarizing their lesson. Their paper includes reflections on their perceptions of how they saw their lesson being played out. For this they draw on informal feedback from their peers and the instructor, which is elicited at the end of each presentation. Note that the primary purpose of the presentations is for students to have an opportunity to practice their public speaking skills. The grade for this written paper represents 20% of their grade for the course.

The games that students in the history of mathematics course used in their lessons during Spring 2008 and Spring 2009 took on a variety of forms. Several students simply posed problems to their peers, and rewarded those individuals or teams who completed the problems the quickest. Rewards generally consisted of a piece of candy or verbal praise. Other students designed games that are adaptations of popular commercially available board games or television game shows. The kind of questions that students chose is what made them appropriate for the history of mathematics class.

Most students who used a game in their lesson incorporated this at the end of their lesson. These students began their lesson by reviewing key concepts from their prior presentation, including the historical and cultural influences on the development of their mathematical topic, and the life of people who played a major role in developing the mathematics. After this review the presenter generally provided some problems to practice the content covered, and then played a game at the end of class, if time allowed. One exception was a student who used a game as a warm-up activity during his lesson.

In the sections that follow we provide several examples of students using games in their lessons. All names are pseudonyms.

Example 1. Laurel's assigned mathematical topic was fractions. As part of the lesson she presented to the class, she engaged her peers in a game of BINGO. Laurel downloaded from the Internet the template for the BINGO cards [5]. She wrote "answers" in each blank space of the grid (see Figure 1).

To play the game, Laurel wrote mathematics problems, one at a time, on the chalkboard, reading them aloud as she wrote. Three of the problems are as follows:

(1) $4\frac{2}{3} + 7\frac{1}{9}$



Figure 1. Sample BINGO board © 2002 EducationWorld, http://www.educationworld.com/a_lesson/TM/WS_lp268-04.html. Reprinted with permission of EducationWorld.

- (2) Rewrite $\frac{31}{8}$ as a mixed number
- (3) Write as a simple fraction: $\frac{3}{4} \div \frac{9}{8} \div \frac{1}{2}$.

Laurel's peers were to complete each arithmetic problem and place a marker on the corresponding box on their BINGO board to indicate the solution. The

first player to get five markers in a row (horizontally, vertically, or diagonally) was the winner. Laurel’s written summary of her lesson included the following self-assessment.

My goal in playing this game was to relate the topic to a fun game that everyone knows. I want my students to see that math can be fun. . . . Fractions seem like an easy concept that each student, especially in college, should understand inside and out. In my observations at a nearby high school for an education class, I observed the exact opposite. The students were in ninth grade and were struggling with simple operations on fractions. This supports my argument that fractions are important in all grades and need to be reviewed after we learn them in elementary or middle school.

Example 2. Dorothy’s assigned mathematical topic was whole numbers, focusing on writing whole numbers in various numeration systems. These systems include the tally system, the Egyptian hieroglyphic numeration system, the Babylonian numeration system (based on two wedge-shaped symbols), the Mayan system (consisting of dots and bars), the Roman numeral system, and the Hindu-Arabic system.

During her lesson, Dorothy divided the class into teams. A representative of each team went to the board and solved a conversion problem. The persons at the board received assistance from their peers, who simultaneously solved the problem at their desks. The first set of problems involved converting to or from the Hindu-Arabic system. For instance, one problem involved converting 1213 to the Egyptian system, and another problem involved converting the numeral represented by the Babylonian symbols $\nabla \nabla \triangleleft \nabla \triangleleft \triangleleft \nabla \nabla \nabla$ to the equivalent Hindu-Arabic numeral. The second set of problems involved converting from one numeral system to another. For instance, students were asked to provide the Babylonian equivalent for the set of symbols given in Mayan notation, as shown in Figure 2.



Figure 2. Mayan notation.

Dorothy hung posters on the front board to assist the teams in their conversions. The posters illustrated the various numeration systems. Students were permitted to use calculators during this game, and for each problem, the team who wrote the correct solution to the conversion problem on the board received a piece of candy.

All students actively participated in this game. The people at their desks solved the problems and made suggestions to their teammates at the board, sometimes overly enthusiastically.

Example 3. Michael's assigned mathematical topic was negative numbers. During his lesson he used an adaptation of a game called *Integer Flash* [1]. For this game students are in groups of three. Each group is provided with a deck of playing cards and a die. Number cards have their face value, and jokers = 0, aces = 1, jacks = 11, queens = 12, kings = 13. Red cards represent negative values and black cards represent positive values. The goal of the game is to be the fastest person to solve integer computation problems. In each group the deck of cards is shuffled and evenly split between the group members. Players keep their cards facedown in a pack. The game begins by rolling the die twice. The result of the roll determines the operations to be performed, with 1 = addition, 2 = subtraction, 3 = multiplication, 4 = division. (If a 5 or 6 is rolled, the die is rolled again.) Each player simultaneously flips over one card from his or her pack, and the group members use the cards, together with the operations, to perform the computation clockwise. The first person to correctly complete the computation wins that round. The person who wins the most rounds is the winner. The original version of the game involved students being grouped into pairs, and performing an integer computation involving just two numbers and one operation. In his written paper, Michael noted that he adapted this game by having students grouped into triads, and performing two operations with three cards because "I felt it would be too simple for college students to simply perform one computation with integers."

Example 4. Adaptations of the game Jeopardy were used frequently. The other students enjoyed competing either individually or in teams in such presentations.

4a. Heini was an atypical student in the history of mathematics class. Instead of being a prospective secondary mathematics teacher, he plans to attend medical school upon graduation. His assigned topic was electronic computers. As part of his presentation, Heini engaged the class in a game he created called "Computer Jeopardy." Using a computer, he projected the following Jeopardy board onto the screen (see Figure 3).

Students in the class were divided into two teams. Teams took turns selecting a category and dollar amount. Corresponding to each, Heini selected the appropriate cell on the board and the computer program displayed a question. If students supplied the correct answer, that team received the points. For

INVENTORS	INVENTION/ CONTRIBUTION	BINARY ARITIMETIC	RANDOM
100	100	100	100
200	200	200	200
600	600	600	600
800	800	800	800
1000	1000	1000	1000

Figure 3. Jeopardy board.

instance, the 200-point question for the category Binary Arithmetic was: *Convert 75 (decimal) to binary*. The category Heini labeled “Random” included questions about his own personal history. This assessed the extent to which his peers had come to know him throughout the semester. Questions in this category included the following: *Where am I originally from?* and *What is my favorite baseball team?*

4b. Maura used a computer version of Jeopardy done in PowerPoint. Her topic was the history of cubic equations. The categories for her game were The Binomial Theorem, Chinese Cube Root, Indian Cube Root, Conic Sections, and Algebra. Before playing the game, when she presented her topic in class, she illustrated the Chinese method with the image of a cube broken into eight parts (see Figure 4).

During the Jeopardy game, two of the questions in the Chinese Cube Root category were related to this figure: *The binomial expansion breaks apart this shape into 8 pieces*, and *Before being broken up, what is the side of the cube equal to?*

Some of Maura’s questions concerned historical facts, like *Who published the book Arybhatiya?* and *This book is where the Chinese Cube Root Method is found*. For the Final Jeopardy question, Maura asked students to solve the cubic equation $x^3 + 9x - 26 = 0$ using Cardano’s formula.

4c. Monica’s topic was the history of linear equations. The categories for her Jeopardy game were People, Places, and Things; The Method of False Position; The Method of Double False Position; and The Method of Al-jabar. Some of her questions were about facts, such as (1) *The Method of False Position was first developed in this country, as seen in the Rhind Papyrus*

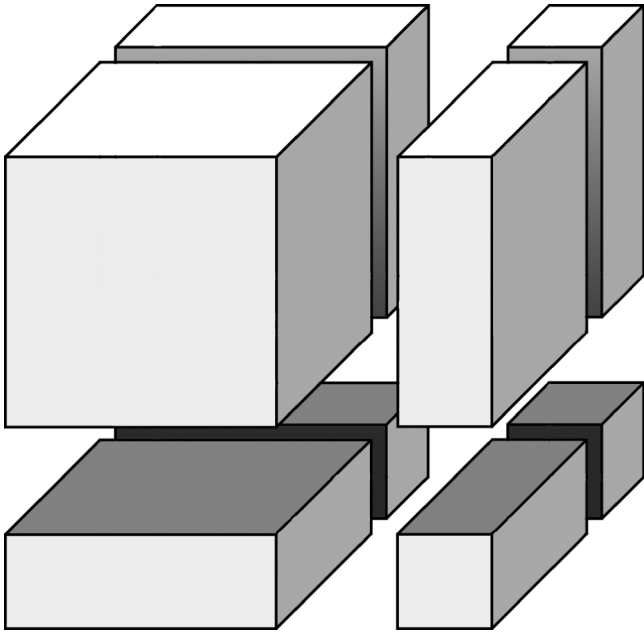


Figure 4. Decomposition of $(r + s)^3$.

(What is Egypt?), and (2) *The name of the influential mathematician who wrote “al-jabar”* (Who is al-Khwarizmi?).

Some of Monica’s other questions required some understanding of the different methods used by mathematicians of different ages and places to solve linear equations. Two examples are as follows: (1) *The first step in solving an equation using the Method of False Position* (What is taking a guess?); and (2) *The factor by which one would multiply an original guess of 7 when solving the following equation using the Method of False Position: $x + (1/7)x = 16$* (What is 2?).

4d. Fanny’s topic was writing algebra with symbols. The categories for her Jeopardy game were Symbol to Person, Person to Symbol, Symbolic Expressions, and Miscellaneous. Her questions were mainly about notations used in different ages, such as the following: (1) *How would Thomas Harriot write $6a^3 + 4b^2 - 5c$? ($6aaa + 4bb - 5c$)*; and (2) *What would this equation be equivalent to today? $Cu.p.7.ce.m.3.co - 4.co.p.2$? ($x^3 + 7x^2 - 3x = 4x + 2$)*.

3. FACTORS FOR SUCCESSFUL USE OF GAMES IN TEACHING THE HISTORY OF MATHEMATICS

Like any instructional method or material, the success of using games to facilitate students’ learning the content depends on how the games are used. The

same principles that make games effective in the learning of mathematics [7] are also valid for games used in the learning of the history of mathematics. In this section we discuss how the games described above satisfy five such principles.

First, each of the games addresses the needs of the class. The history of mathematics is replete with names, facts, and notations that are not easy to remember. By playing the games illustrated above, students practice the names, facts, and notations in a setting they enjoy, rather than trying to memorize by rote. In this way, the games contribute to students' learning history of mathematics facts and ideas in ways more effective than those using other approaches.

Second, the games were used at appropriate times in the course, during a regular class session when the ideas or facts were being taught. Games were relatively short so that students did not lose interest, and all students actively participated in the games.

Third, the games were planned carefully and conducted in an organized way. Each of the games introduced an element of chance to give everybody (not just the more able students) the opportunity to win and be successful. The informality and excitement did not interfere with the purpose of the game. Before the game, students were informed of the purpose of the game, its rules, and the way to play it. The presenters established ground rules so that everybody enjoyed the game. Usually, the loss of points for breaking rules or getting too loud and excited is enough to ensure appropriate behavior.

Fourth, students accepted responsibility for learning from the game. We discussed game results briefly and shared comments after each game. In other settings follow-up activities such as additional readings could be used to emphasize this responsibility. The instructors also evaluated informally to what extent the game was successful in promoting the learning desired. It was interesting to see how much specific information students were able to retrieve, during game playing, from their readings and presentations by their fellow students.

Fifth, each game had a title, rules, and instructions of how to play. In addition, all the necessary materials were included. The games' size was such that they were easily used (commercially available board games are a good practical reference), and were made of sturdy materials with an attractive appearance. Some of the students also took advantage of available computer resources to develop their games. In this case, the computer program was easy to use and reliable.

4. CONCLUSION

In the history of mathematics course offerings during Spring 2008 and Spring 2009, approximately one-third of the students incorporated a game into a portion of their 45-minute mathematics lesson that they presented to their

peers. Students actively and enthusiastically participated in the games during class. They made their best effort to get the right answers and to do well in the games. To make the game more fun, students sometimes included small rewards for correct answers, such as a small piece of candy, or slightly embarrassing punishments for wrong answers such as being voted off their team. Several methods were used to determine assignment of students to groups for playing the games (e.g., counting off). At the beginning of play, students sometimes would play another game (e.g., rock-paper-scissors) to determine the order of play.

Although the class presentations were not formally evaluated, we think the design and use of instructional materials and educational games had several benefits for the majority of the students, and therefore were a good use of class time. Students needed to apply their knowledge of history of mathematics in the design of their games. Students felt motivated and interested in the history of mathematics, and saw how they could integrate it into their own teaching of mathematics. Lastly, they felt better prepared for their future responsibility as teachers in their own classrooms.

These ideas are summarized by Rose, who explained her rationale for designing a game and infusing competition into its execution, all in the context of her future teaching:

Because of the difficulty of the [quadratic problems phrased in historical language], I had the students work in groups, so they could bounce ideas off one another. In this way they could learn from each other, because what one student doesn't understand, another one may. This would not just be group work though, but also a competition between groups. Each group received the same problem, and the first three groups to get an answer would choose a member to put their work on the board. As a class, we would compare answers and strategies, and debate which answer is correct. I chose to introduce the problems this way, instead of doing an example, and then have them copy it down, because I have learned from one of my education courses that giving students a problem they have never seen before, and having them grapple with it, allows them to think critically, and makes them pull from prior knowledge, and try to transfer that knowledge into a new situation. The competition strategy gives the students incentive to want to solve the problem, which I feel is a great way to engage the students.

ACKNOWLEDGMENTS

The authors express appreciation to Dr. Kathy Clark, Florida State University, for helping to conceptualize the development of the history of mathematics course at the University of Delaware during Spring 2008 and 2009.

REFERENCES

1. Beery, J., G. Cochell, C. Dolezal, A. Sauk, and L. Shuey. 2005. Negative numbers. In Katz, V. J., and K. D. Michalowicz (Eds.). *Historical Modules for the Teaching and Learning of Mathematics* [CD-ROM]. Washington, DC: Mathematical Association of America.
2. Bright, G. W., J. H. Harvey, and M. M. Wheeler. 1985. *Learning and Mathematics Games*. Reston, VA: National Council of Teachers of Mathematics.
3. Forman, S., and S. Forman. 2008. Mathingo: Reviewing calculus with bingo games. *PRIMUS*. 18(3): 304–308.
4. Gallegos, I., and A. Flores. 2010. Using student-made games to learn mathematics. *PRIMUS*. 20(5): 405–417.
5. Hopkins, G. 2002. Math bingo. *Education World*. http://www.education-world.com/a_lesson/TM/WS_lp268-04.html. Accessed 20 October 2009.
6. Huntley M. A., and A. Flores. 2010. A history of mathematics course to develop prospective secondary mathematics teachers' knowledge for teaching. *PRIMUS*. 20(7):603–616.
7. Johnson, D. A., and G. R. Rising. 1972. *Guidelines for Teaching Mathematics*, 2nd Edition. Belmont, CA: Wadsworth.
8. Randel, J. M., B. A. Morris, C. D. Wetzel, and B. V. Whitehill. 1992. The effectiveness of games for educational purposes: A review of recent literature. *Simulation and Gaming*. 23: 261–276.

BIOGRAPHICAL SKETCHES

Mary Ann Huntley taught the history of mathematics course at the University of Delaware in 2008. At Cornell University, she is on the faculty in the Mathematics Department. She teaches mathematics courses, and directs mathematics outreach and K-12 education activities. Her research focuses on curricular influences on the teaching and learning of secondary school mathematics.

Alfinio Flores is a professor of mathematics education in the Department of Mathematical Sciences at the University of Delaware, and taught the history of mathematics course in 2009. He teaches mathematics and mathematics education courses *con ganas* to prospective and in-service teachers of mathematics. He uses all kinds of resources, including the history of mathematics, to help make mathematics meaningful for students and teachers.